Pre-requisites: Nil

PH7302E CONDENSED MATTER PHYSICS

L=3 T=1 P=0 O=8 C=4

Total Lecture Sessions: 39

Course Outcomes:

CO1: Differentiate between different lattice types and explain the concepts of reciprocal lattice and crystal

diffraction.

CO2: Distinguish different excitations in crystals and explain the specific heat.

CO3: Apply free electron model and interpret the properties of metals.

CO4: Explain the concept of energy bands and use it to find electrical properties of materials.

CO5: Explain electrical and magnetic properties of materials.

Topics:

Crystal Physics: a brief overview of crystal structure, reciprocal lattice, Bragg's law of diffraction

– Lattice Vibrations and Thermal Properties – Monoatomic and diatomic lattices, normal modes of lattice

vibration, phonons and density of states, dispersion curves – Specific heat: classical, Einstein and Debye

models – Thermal conductivity.

Free Electron theory: Dependence of electron energy on the wave vector, E-k diagram – Free electron theory

of metals- Thermal and Electrical transport properties, Electronic specific heat, Fermi surface – Failures of

free electron theory – Bloch theorem – Nearly free electron model and origin of energy bands.

Energy Band Theory: Kronig-Penney Model, tight binding approximation – Motion of electrons in bands and

effective mass – Semiconductors: Intrinsic and extrinsic semiconductors, Charge carrier density in intrinsic

semiconductors, Doping, Carrier densities in doped semiconductors.

Dielectric properties of insulators – Pizoelectric and Ferroelectric materials – Optical properties of solids –

Magnetism-exchange interaction, diamagnetism, paramagnetism, ferromagnetism and antiferromagnetism.

References:

1.C. Kittel, Introduction to Solid State Physics, Wiley, 2007.

2.N. D. Mermin and N. Ashcroft, Solid state Physics, Thomson, 2007.

3.H. Iback and H. Lüth, Solid State Physics, Springer, 2009.

4.S. Elliot, The Physics and Chemistry of Solids, John Wiley & Sons, 1998.

5.A. Omar, Elementary Solid State Physics, Addison-Wesley , 2005.